

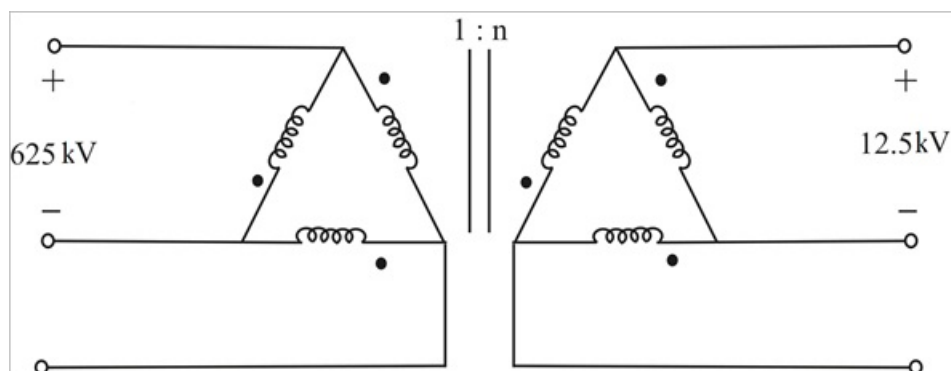
Problem

A three-phase Δ - Δ transformer is used to step down a line voltage of 625 kV, to supply a plant operating at a line voltage of 12.5 kV. The plant draws 40 MW with a lagging power factor of 85 percent. Find: (a) the current drawn by the plant, (b) the turns ratio, (c) the current on the primary side of the transformer, and (d) the load carried by each transformer.

Step-by-step solution

Step 1 of 7

The three phase Δ - Δ transformer is,



Step 2 of 7

Given that,

$$P_T = 40 \text{ MW}$$

$$\cos \theta = 85\%$$

$$\cos \theta = 0.85$$

Step 3 of 7

(a) For the Δ - Δ three phase transformer the real power is (S_T)

$$P_T = \sqrt{3} V_L I_L \cos \theta$$

$$40 \times 10^6 = \sqrt{3} \times 12.5 \times 10^3 \times I_L \times 0.85$$

$$\therefore I_L = 2.1736 \text{ kA}$$

Therefore, the secondary line current is 2.1736 kA

Step 4 of 7

(b) Turns ratio $(n) = \frac{V_{LS}}{V_{LP}}$

$$n = \frac{12.5 \times 10^3}{625 \times 10^3}$$

$$\therefore \boxed{n = 0.02}$$

Step 5 of 7

(c) Primary current $(I_{LP}) = I_{LS} \times n$

$$I_{LP} = 2.1736 \times 10^3 \times 0.02$$

$$\therefore \boxed{I_{LP} = 43.47 \text{ A}}$$

Step 6 of 7

(d) $S_T = \sqrt{3} V_L I_L$

$$S_T = \sqrt{3} \times 12.5 \times 10^3 \times 2.1736 \times 10^3$$

$$\therefore S_T = 47.06 \text{ MVA}$$

Step 7 of 7

Therefore , the load carried by each transformer is,

$$S_{\text{each}} = \frac{47.06 \times 10^6}{3}$$

$$\therefore \boxed{S_{\text{each}} = 15.69 \text{ M VA}}$$